

DD-KOOPMAN: EXACTNESS ON A BUDGET FOR COMPACT MULTI-STEP CSI PREDICTION

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ABSTRACT

Multi-step channel state information (CSI) prediction is usually cast as sequence-to-sequence learning, yet a specular channel’s future is fixed by a few parameters: state estimation followed by a small linear extrapolation, and the budget belongs in the estimator. DD-Koopman builds that split – input-side estimation with a conditional spectral readout over a parameter-free delay–Doppler lift – in 173,190 parameters, reaches 0.1402 mean NMSE against the 21.9M-parameter reference’s 0.1246 at $126.5\times$ fewer parameters and $4.98\times$ lower latency, and improves on a 432-setting unseen-propagation slice across three seeds. Exact prediction of a K -mode exponential support requires $PV = Q$: at full grid support the exact transfer is unique with zero degrees of freedom and reduces to periodic continuation, while covering K grid modes buys $2T(N-K)$ degrees of freedom at a minimum white-noise gain of KT/N ; off grid, the worst-case mixture error is the budget times the spectral defect, certified from a finite scan. The measurements close the loop: the zero-parameter exact transfer scores 3.2883 against the learned model’s 0.1402, a fitted readout on the same 16 grid bins removes most of that error, and the largest measured degradation is estimator-side, not transfer-side.

Index Terms— Channel prediction, CSI, operator learning, delay–Doppler, parameter efficiency

1. INTRODUCTION

The CSI prediction task is to forecast the next four channel states from sixteen historical observations on a $4\times 4\times 2$ MIMO array over 300 subcarriers. The CSI-4CAST benchmark [1] standardises the splits and the metric and publishes a reference model that reaches a mean normalised mean square error (NMSE) of 0.1246 on its official 162-setting TDD-regular grid, at the cost of 21,913,750 parameters. Our harness reproduces the published number within 0.105% (the benchmark does not seed its RNG), so every comparison below runs on the same protocol.

Per-path frequency, radius and delay amplitude fix a specular channel’s future, so the task is short-window identification and extrapolation rather than sequence modelling. The analysis below characterizes which transfers are exact, what full-grid periodic exactness costs on this benchmark, and how badly a transfer degrades off its support; the experiments locate the practical bottleneck: the exact transfer itself scores 3.2883 against the learned model’s 0.1402, and removing the input-side denoiser costs 27.3% NMSE.

Our contributions are:

1. **Exactness as a budget.** We characterize finite-window spectral extrapolation as a constrained problem: exact prediction of a K -mode exponential support requires $PV = Q$ – unique with zero degrees of freedom at full grid support, $2T(N-K)$ real degrees of freedom at K modes with the minimum white-noise gain KT/N attained by the minimum-norm member $P_0 = QV^\dagger$ – and over a compact off-grid class the worst-

case noiseless error is the amplitude budget times the spectral defect, certifiable from a finite scan.

2. **A predictor built to the split.** DD-Koopman separates estimation from extrapolation: a 12,768-parameter conditioned spectral operator carries 7.4% of the 173,190 parameters, the rest in input processing. The split is testable, and the tests agree: a grid-pinned retrain matches the reported model, so off-grid pole placement is not necessary once the estimator and readout are re-optimized; removal costs under matched 240-epoch controls rank the roles – 0.8% for pole conditioning, 14.3% for the readouts’ observation conditioning, 27.3% for the input denoiser.
3. **Design dominates capacity, and the analysis is checkable.** Accuracy stays within 0.1367–0.1423 across a $5.6\times$ parameter scan (58k–327k) while comparable-capacity architectures remain above 0.18; on the frozen checkpoints the transfer expansion reproduces the network output exactly and the finite-scan defect certificate holds on every evaluated instance. DD-Koopman reaches 0.1402 against the published 0.1246 with $126.5\times$ fewer parameters and $4.98\times$ lower measured latency, and improves mean NMSE on a 432-setting unseen-propagation slice, each paired CI entirely below zero (§4.5, §4.6).

2. RELATED WORK

Benchmark and published baselines. CSI-4CAST [1] provides the training grid, the three test splits, the harness, and the published reference rows we compare against, including the language-model predictor LLM4CP [2]; neural channel prediction predates the benchmark [3, 4, 5]. Cross-paper NMSE is not comparable, so we use the benchmark’s published reference rows and re-implement the Table 2 models through its harness.

Delay-domain representations. OTFS [6] made the delay–Doppler plane the modulation domain on the same sparsity. ChannelKAN [7], MambaCSP [8], PCGAE-Net [9] and CSI-BERT2 [10] embed delay features next to a generic model. We instead study the extrapolation map itself: its exact solution set, its noise amplification, and its off-support spectral defect, with the plane as the space where the operator is diagonal.

Dynamics-based and state-space prediction. The Koopman programme lifts nonlinear dynamics to linear evolution of observables [11, 12, 13], approximated by EDMD [14], deep embeddings [15] and neural operators [16, 17]. Structured, selective and gated recurrences [18, 19, 20, 21], with CSI adaptations [8, 22], learn input-dependent recurrences in latent coordinates. Our operator is diagonal in an explicit delay–Doppler basis with a closed-form transfer, and we measure what one frozen model retains under shift.

Efficient forecasting backbones. PatchTST [23], iTransformer [24], TimeMixer [25], TSMixer [26] and Mamba [20] compactify generic sequence modelling; we port, retrain and score

them under the budgets stated in Table 2, alongside the attention, recurrent and linear families they grew from [27, 28, 29, 30].

3. METHOD

Lift. Per antenna, the history is $\mathbf{H} \in \mathbb{C}^{16 \times 300}$ (16 slots, 300 sub-carriers) and the target is the next four slots; Fig. 1 gives the signal path. Two parameter-free transforms map the observation into the working basis: $a = \text{IDFT}_{\text{sub}}(\mathbf{H})$ converts each slot to its delay profile (300 taps), and $\mathbf{D} = \text{DFT}_{\text{time}}(a)$ converts along time, giving the delay–Doppler plane. A specular channel concentrates in a few delay taps and few Doppler modes, so the dynamics live in a sparse, physically derived coordinate system; the lift has no parameters, and all 173,190 parameters belong to the operator, the residual denoiser, and the gated correction paths.

Input processing (the estimator). A small convolutional *residual denoiser* acts on the input history before the lift through a zero-initialised head, starting as the identity. Two residual paths then re-enter the *same* operator, each contributing a gated difference of two operator evaluations, $\hat{y} = \text{base} + g_1(\text{op}(\text{mix}(x)) - \text{base}) + g_2(\text{op}(\text{res}(x)) - \text{base})$: *mix* applies per-tap adaptive reweighting (initialised to one) and a temporal convolution (initialised to zero); *res* is the unexplained part of a learned delay-profile split, a delay-domain analogue of series decomposition [31]. Their scalar gates start at zero, so the sum initialises to exactly the base; each antenna is processed independently with shared weights. The parameter inventory in §4.2 confirms the estimator split.

Conditioned spectral transfer. For horizon $h \in \{1, \dots, 4\}$ (target slot $15+h$), the base prediction extrapolates each mode exponentially:

$$\hat{a}[h, l] = \sum_m G_h[m](\mathbf{c}) \mathbf{D}[m, l] \zeta_m(\mathbf{c})^h, \quad (1)$$

where m indexes the 16 Doppler bins, l the 300 delay taps, and applying DFT_{sub} to $\hat{a}$ reconstructs the predicted frequency-domain CSI. The poles are $\zeta_m(\mathbf{c}) = r_m(\mathbf{c}) e^{j2\pi(m+\varepsilon_m(\mathbf{c}))/16}$ with total offset $\varepsilon_m(\mathbf{c}) = \delta_m + \Delta_m(\mathbf{c})$: a learned static offset δ_m plus an observation-conditioned shift $\Delta_m(\mathbf{c})$, each in $(-\frac{1}{2}, \frac{1}{2})$, so $\varepsilon_m(\mathbf{c})$ places every pole continuously between the DFT bins. Both the radii $r_m(\mathbf{c}) \in (0, 1)$ and the offsets depend on the observation through a context $\mathbf{c}$ that summarises the observed Doppler power averaged over delay taps; the core $K_c = \text{diag}(\zeta_m(\mathbf{c}))$ is held fixed across the four horizons. The gains $G_h[m](\mathbf{c})$ are complex, indexed by horizon and mode, shared across delay taps, and modulated per observation from the same context; they calibrate the finite-window readout, while the semigroup law $K_c^{h+k} = K_c^h K_c^k$, non-expansive in every power ($\|K_c^h\|_2 \leq 1$), belongs to K_c alone. The reported model initializes its offsets on the grid ($\delta_m, \Delta_m(\mathbf{c})$ start at zero) and applies the ridge least-squares warm start.

Effective transfer and notation. $N=16$ history slots, $T=4$ horizons, $L=300$ delay taps; $X \in \mathbb{C}^{N \times L}$ and $Y, \hat{Y} \in \mathbb{C}^{T \times L}$ in delay coordinates; $P \in \mathbb{C}^{T \times N}$ an effective temporal transfer; z an exponential factor with atoms $v_N(z) = (1, z, \dots, z^{N-1})^T$, $u_T(z) = (z^N, \dots, z^{N+T-1})^T$; $\mathcal{Z}$ a designated factor set; $\dagger$ the pseudoinverse, $\mathbf{H}$ the conjugate transpose. All branches re-enter one operator, so each forward pass acts on its branch input through the same conditional form: $P_j = W_j F_N$ with $(W_j)_{hm} = G_{hm}(\mathbf{c}) \zeta_m(\mathbf{c})^h$, and with gates $\alpha = (1-g_1-g_2, g_1, g_2)$,

$$\hat{Y} = P_X X + R_X, \quad P_X = \sum_j \alpha_j P_j, \quad R_X = \sum_j \alpha_j P_j (A_j - X), \quad (2)$$

holds exactly, both objects computed from the same pass. Here A_0, A_1, A_2 are the delay-domain histories reaching the base,

reweighting/mixer, and decomposition-residual branches, and each P_j is recomputed from its branch context, $P_j = P(A_j)$: the denoiser, reweighting and decomposition shape both P_X and R_X . P_X carries $T \times N$ complex coefficients, independent of the parameter count.

Proposition 1 (exactness and its price). For $K \leq N$ distinct factors, let $V = [v_N(z_k)]$, $Q = [u_T(z_k)]$. A fixed P predicts every pair $VB \mapsto QB$ exactly iff $PV = Q$; the solution set has real dimension $2T(N-K)$, and $P_0 = QV^\dagger$ is its unique minimum-Frobenius-norm member. At full grid support ($K=N$ unit roots) $Q = [I_T \ 0]V$, so the solution is unique with zero degrees of freedom and $P_0 = [I_T \ 0]$: periodic continuation, $\hat{Y} = X[0:T]$. On a grid support the output-noise gain $G(P) = \mathbb{E} \|Pe\|_2^2 / \sigma^2 = \|P\|_F^2$ for $e \sim \mathcal{CN}(0, \sigma^2 I_N)$ is minimized among exact transfers by P_0 , with $G(P_0) = KT/N$; off grid $\text{tr}[Q(V^H V)^{-1} Q^H]$ replaces it [32].

Theorem 1 (spectral-defect bound). Let $d_P(z) = P v_N(z) - u_T(z)$ and $\delta_Z(P) = \max_{z \in \mathcal{Z}} \|d_P(z)\|_2$ over compact $\mathcal{Z}$. Over finite mixtures $S = \sum_k v_N(z_k) b_k^T$, $Y = \sum_k u_T(z_k) b_k^T$ with $z_k \in \mathcal{Z}$ and $\sum_k \|b_k\|_2 \leq M$, the worst noiseless error of a fixed P is exactly $M \delta_Z(P)$, attained at a single atom; and the implemented predictor, with P_X and R_X fixed by the observed X , obeys

$$\|\hat{Y} - Y\|_F \leq M \delta_Z(P_X) + \epsilon \|P_X\|_2 + \|R_X\|_F \quad (3)$$

whenever $\|X - S\|_F \leq \epsilon$: a per-input diagnostic separating model-class defect, input perturbation, and correction magnitude; which modules help is established by the matched ablations of §4.4.

Corollary 1 (finite-scan certificate). Let $\mathcal{G}$ be finite with $\mathcal{Z} \cup \mathcal{G} \subset \{z : |z| \leq \rho\}$ and every $z \in \mathcal{Z}$ lie within η of some $g \in \mathcal{G}$. With $L_v(\rho) = (\sum_{p=1}^{N-1} \rho^{2p-2})^{1/2}$ and $L_u(\rho) = (\sum_{p=N}^{N+T-1} \rho^{2p-2})^{1/2}$, $\delta_Z(P) \leq \hat{\delta}_{\mathcal{G}}(P) + \eta(\|P\|_2 L_v(\rho) + L_u(\rho))$, where $\hat{\delta}_{\mathcal{G}}$ is the grid maximum: a finite scan certifies the continuous defect.

Proof sketch. (2) adds and subtracts X branchwise; V has full column rank (Vandermonde), so every exact P is $P_0 + Z(I - VV^\dagger)$, orthogonal to P_0 ($P_0(I - VV^\dagger) = 0$) with arbitrary Z ; the bound expands the mixture into outer products and attains it at one atom; the certificate uses $|z^p - g^p| \leq p\rho^{p-1}|z - g|$. ■

On the frozen checkpoints these objects are measurable: the expansion reproduces the network’s delay output to floating-point precision (relative residual 3×10^{-15} in float64; the float32 residual 1.97×10^{-6} is the model’s own forward arithmetic). The implementation satisfies the finite-scan inequality in all 311,040 evaluated instances – 257 unit-circle angles, $\rho=1$, $\eta=\pi/256$; the mixture bound (3) holds with zero violations in 10,800 synthetic trials, with median bound-to-true ratio $3.84 \times$ (5th–95th percentile 2.37–6.24).

4. EXPERIMENTS

4.1. Setup

The training grid has 27 subsets (3 channel models $\times$ 3 delay spreads $\times$ 3 speeds), 24,300 training and 2,700 validation samples. The test splits are the *regular* split (162 settings: channel model $\times$ delay spread $\times$ speed $\times$ six SNRs) and the *robustness* split (486 settings: phase/burst noise and packet drops over the 27 regular combinations). The *generalization* slice has 432 settings over 72 combinations drawn from channel models B and E, with delay spreads 50/200/400 ns and speeds 3–42 m/s, all absent from training; the metric is the benchmark’s mean NMSE over the four horizons. For paired statistics we average NMSE over the six SNRs in each channel–delay–speed cell and bootstrap paired cell differences (5,000 resamples, 95% CI); intervals describe variation across cells. Training uses batch 32, Adam [33] at $5 \cdot 10^{-4}$ with a plateau scheduler and early stopping (600 epochs reported, 240 for the ablation table), three seeds.

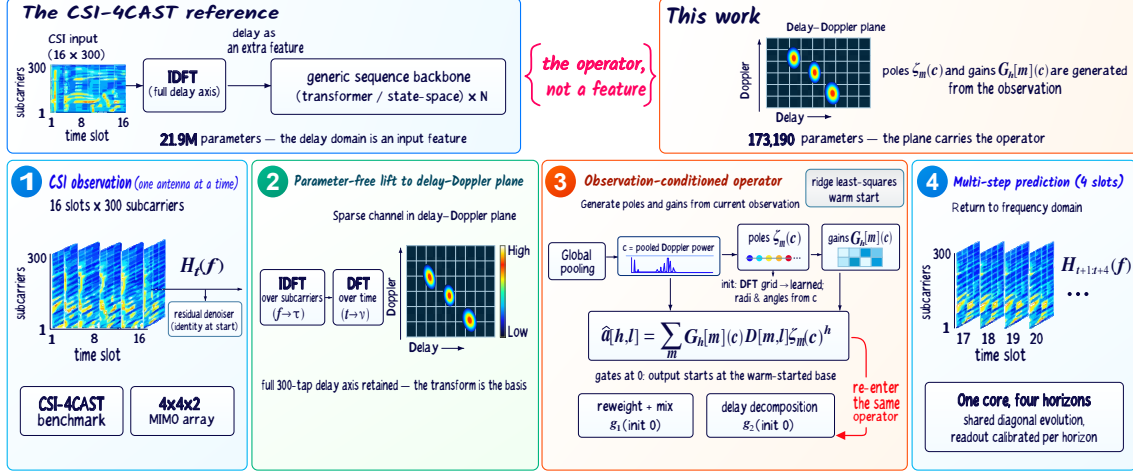


Fig. 1. The 21.9M-parameter CSI-4CAST reference attaches the delay domain to a generic backbone; this work puts the evolution operator in that plane (173,190 parameters): (1) residual denoiser on the observed history; (2) parameter-free IDFT/DFT lift to the delay–Doppler plane; (3) pooled context generating the poles $\zeta_m(c)$ and gains $G_h[m](c)$, two residual paths gated from zero re-entering it; (4) four-slot prediction.

4.2. Where the parameters go

The implementation follows the split: the conditioned spectral operator holds 12,768 parameters (7.4%), per-tap reweighting 154,328 (89.1%), the temporal mixer 4,200, the residual denoiser 1,590, the delay decomposition 300, and four scalar gates, for 173,190 in total. Table 2 places this split on the accuracy–efficiency plane.

4.3. The price of exactness

Proposition 1’s full-grid exact transfer scores end to end without a fitted parameter: it is periodic continuation, reaching 3.2883 on the regular split – the worst row in Table 2 – and 2.7558 and 2.7518 on the shifted splits. Raising the support from the adaptive top-1 bin (1.3771) to the full grid degrades monotonically across $K \in \{1, 2, 4, 8, 16\}$, consistent with truncation suppressing noisy modes. Staying inside the same 16-bin representation but replacing periodic exactness by a ridge least-squares fit of the same 4×16 gains – the model’s own warm start, a 256-sample fit – reaches 1.2504 (1.1366 and 1.4767 shifted), and fitting on all 24,300 training samples reaches 0.8173: within this fixed-grid comparison, the exactness constraint accounts for most of the gap to the fitted readout; the ablations below assess the rest. The harness’s $K=16$ transfer matches the analytic continuation to float32 precision (1.7×10^{-6}).

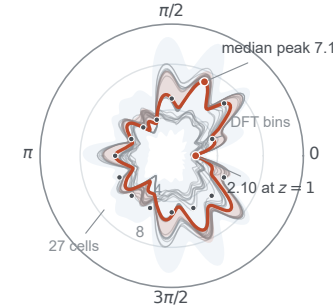


Fig. 2. Off-grid response of the learned transfer (seed 42, validation split): radius shows the per-angle median defect $\|d_P(z)\|_2$, grey per-cell medians, blue 5th–95th percentile bands, dots the 16 DFT bin phases – reference phases, not zero-defect constraints.

Pinning the poles to the grid and retraining recovers the reported model: 0.1401 against 0.1402 on the regular split and 0.2688 against 0.2716 on the generalization slice (three seeds), and the fix holds on the 486-setting robustness split (0.1775 against 0.1761). Off-grid placement is thus not necessary after re-optimization; the fitted checkpoints still use the freedom, with 89% of trained poles more than

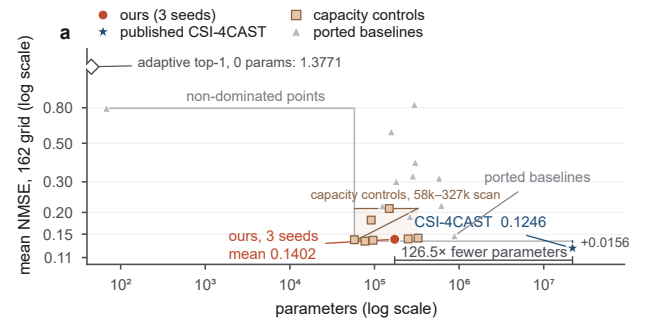


Fig. 3. The accuracy–efficiency plane; the reported model is the three-seed-validated operating point. The star marks the published model, dots the three seeds, squares the capacity controls, triangles the baselines, and the step line the non-dominated points.

0.05 bins off grid and all 16 modes moving across scenes. Figure 2 measures the learned transfer’s off-grid response over the unit circle: the defect falls to 2.10 at $z=1$, peaks at 7.1 in the mid-band, and its per-cell peak lies between 4.1 and 8.3 in every one of the 27 propagation cells: the median defect stays elevated over a broad mid-band rather than being driven by a single angle.

Snapping the offsets of an already-fitted checkpoint, every weight frozen, moves the fitted transfer outside its calibrated region. Setting $\varepsilon_m(c) = 0$ degrades the official grid from 0.1385 to 0.1923 (+38.8%) with all 162 settings worse, every constant global offset is worse still, so the gain is per-mode, per-sample placement, and the penalty rises monotonically over horizons. The same intervention degrades all 1,728 settings across the four splits (+17.7% to +38.8%), and on every shifted slice it pushes the operator below the published model (+0.0288 to +0.0691, all CIs entirely above zero).

4.4. Ablations: the estimator dominates

Module factorial. Table 1 runs the full 2^3 loss factorial. The residual denoiser is the dominant single effect (+31.7% average main effect across the four flip pairs), and removing the denoiser together with either other module costs more than the sum of the parts: the components complement rather than duplicate. A squared FreDF-style spectral term [34] reweights the same regression by Parseval’s iden-

Table 1. Loss-off factorial on the official grid (spectral loss off, 240 epochs, seed 42; ✓ = active, off = removed). Deltas are computed from the tabulated NMSEs.

denoiser	decomp.	poles	NMSE	Δ	$\Delta\%$
✓	✓	✓	0.1397	—	—
✓	✓	off	0.1408	+0.0011	+0.8%
✓	off	✓	0.1425	+0.0028	+2.0%
✓	off	off	0.1458	+0.0061	+4.4%
off	✓	✓	0.1779	+0.0382	+27.3%
off	✓	off	0.1853	+0.0456	+32.6%
off	off	✓	0.1862	+0.0465	+33.3%
off	off	off	0.1968	+0.0571	+40.9%

Table 2. Mean NMSE on the official 162-setting grid. Published rows (including the classical predictors) come from the benchmark’s reference table; all other rows were trained, or for the DFT-grid extrapolators fitted without training, and scored through the same harness. $\times$ is the ratio to the published row; Budget is the training-epoch cap.

Model	Params	Budget	NMSE	$\times$
CSI-4CAST (published) [1]	21,913,750	—	0.1246	1.00
LLM4CP (published) [2]	—	—	0.1413	1.13
Adaptive top-1 DFT-grid extrap.	0	—	1.3771	11.05
Full-grid exact extrap.	0	—	3.2883	26.39
ours, mean of 3 seeds	173,190	600 ep	0.1402	1.13
ours, fixed-grid retrained (3 seeds)	173,190	600 ep	0.1401	1.12
MambaCSP (port, plateau)	881,604	100 ep	0.1468	1.18
CSI-4CAST shrunk (91k)	91,410	240 ep	0.1807	1.45
capacity-matched TF	149,560	240 ep	0.2106	1.69
TSMixer-L	124,337	240 ep	0.2177	1.75
PatchTST	616,560	240 ep	0.2190	1.76

Re-implemented rows are single seed, early-stopped under the shared validation rule; the fixed-grid retrained row pins its offsets to the DFT grid (three seeds, §4.3); the DFT-grid rows are fitted without training – top-1 continues the strongest grid bin periodically, and the full-grid row is Proposition 1’s exact transfer at $K=N$ (§4.3).

tity, yet every matched comparison trains worse with it (two seeds at 600 epochs, one at 240); we drop it.

On the generalization slice the same factorial attributes the shift advantage to the residual denoiser (+20.9% without it, 95% CI [+0.0431, +0.0713]) and secondarily to the delay decomposition (+4.0%); removing the pole conditioning slightly helps (−0.9%).

Gain conditioning. Freezing the readout gains’ per-observation modulation to per-mode constants, under the factorial protocol (240 epochs, seed 42), costs 0.1584 against 0.1386 for a same-environment control arm – +14.3%, an order of magnitude above pole conditioning (+0.8%); the control lands next to the factorial reference 0.1397, so the gap is the intervention. Under these matched controls, gain conditioning carries the larger marginal effect; the frozen-checkpoint snap above shows the trained poles still matter to their jointly fitted configuration.

The transfer applied at each pass is only $T \times N$; the hard part is inferring it from a noisy finite history. The tested bottleneck sits there: input processing costs 27.3% when removed, gain conditioning 14.3%, pole conditioning 0.8%, and a $5.6\times$ parameter scan moves accuracy by less than 0.006.

4.5. Accuracy–efficiency operating point

Shrinking the published architecture to 91,410 parameters yields 0.1807 at convergence, still $1.29\times$ our mean; a transformer sized to match our parameters reaches 0.2106. The MambaCSP port (100 epochs, validation plateau) remains behind the reported model on

all three splits (0.1468, 0.2796 and 0.1805) at $5.1\times$ our parameters; the remaining ports trail further (GRU 0.3239, TCN 0.8353, DLinear 0.7941, MLP 0.3143, Mamba 0.3011). Within our own architecture, accuracy is width-insensitive (0.1367–0.1423 across a $5.6\times$, 58k–327k parameter scan), while halving the training windows costs about 17% NMSE: over the tested range, performance is more sensitive to data than to width.

Table 2 and Fig. 3 give the trade-off: our three-seed mean NMSE is 0.1402 against the published model’s 0.1246 (the seeds score 0.1385, 0.1442 and 0.1378); the per-seed paired differences over the 27 channel cells are +0.0139, +0.0196 and +0.0132, each CI entirely above zero.

The classical predictors bound what fixed assumptions deliver: path-delay collapses from 0.2300 to 0.3231 under shift, autoregressive (0.7874) and Wiener (0.8236) degrade to 1.0469 and 1.0862 against the released checkpoint’s 0.2872, and the adaptive top-1 DFT-grid extrapolator reaches 1.3771, 1.2465 and 1.2790.

The reported model carries 173,190 parameters and 6,514,464 MACs per antenna sample against the published 21,913,750 and 804,901,376: $126.5\times$ and $123.6\times$ ours, counted over matmul and convolution only; the parameter-free DFT/IDFT lift and elementwise products are excluded. Forward latency (RTX 4090 D, batch 32, fp32, 12 repeats) is 2.24 ± 0.05 ms against 11.15 ± 0.02 ms per scene: $4.98\times$ faster. With MRT beamforming the released checkpoint keeps more of the perfect-CSI rate (2.06% vs. 2.70% loss, paired +0.64 pp).

4.6. Behaviour under distribution shift

On the two shifted splits the three frozen checkpoints hold: the 432-setting generalization slice gives 0.2685, 0.2767 and 0.2695 against the released checkpoint’s 0.2872. Each paired difference has a 95% CI entirely below zero (−0.0187, −0.0105, −0.0177), and the three-seed mean difference is −0.0156 [−0.0250, −0.0070] over 72 channel cells (5,000-resample bootstrap). The operator leads on model B (−0.0400, 95% CI [−0.0531, −0.0271]) while on model E the released checkpoint stays ahead (+0.0088 [+0.0046, +0.0132]) at an order of magnitude smaller NMSE. Mobility holds: below 10 m/s the three-seed difference is −0.0204 [−0.0378, −0.0057]; above 30 m/s they are indistinguishable (+0.0001, CI [−0.0180, +0.0169]).

The shrunk architecture (0.3039), capacity-matched transformer (0.3427) and with-loss variant (0.2951) all trail on this slice; the MambaCSP port (0.2796) is not separated (CI [−0.0178, +0.0019]).

Across the 486-setting robustness split the released checkpoint leads: 0.1623 against 0.1772, 0.1735 and 0.1777.

On a second shift axis (A, C and D outside their training delay-spread and speed ranges; TDD only) the operator stays ahead on the mean (0.2031 against 0.2182, −0.0151, 95% CI [−0.0256, −0.0054]), with the gain concentrated at 400 ns. Consistent with the estimator-based interpretation, the advantage is largest where the pooled Doppler context averages over many taps rather than few; the same holds on the independent B/E axis (−0.0367 at 400 ns, CI [−0.0549, −0.0189]; absent at 50 ns).

5. CONCLUSION

A multi-step CSI predictor is an estimator followed by a small linear transfer, and exactness in that transfer is a constrained resource. On this benchmark, full-grid periodic exactness scores 3.2883 against the learned model’s 0.1402 and a fitted readout on the same 16 bins removes most of that error; removal costs rank 27.3% (input denoiser), 14.3% (gains) and 0.8% (poles) under matched controls. DD-Koopman holds a three-seed unseen-propagation advantage at $126.5\times$ fewer parameters and $4.98\times$ lower latency.

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